Lecture 10

Tuesday, 15 September 2026 7:22 PM

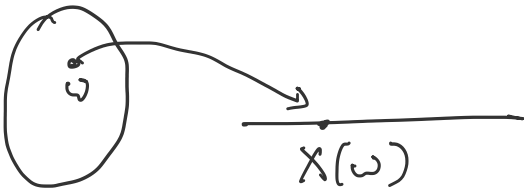
Random Vectors (Multivariate distributions)

Random Variable $(\Omega, \mathcal{F}, \mathbb{P})$

CDF.

Discrete → PMF

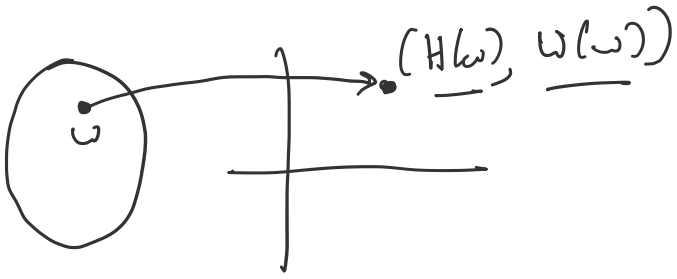
Conts → PDF



Random vector → Randomly sample from the class.

(Height, Weight, age)

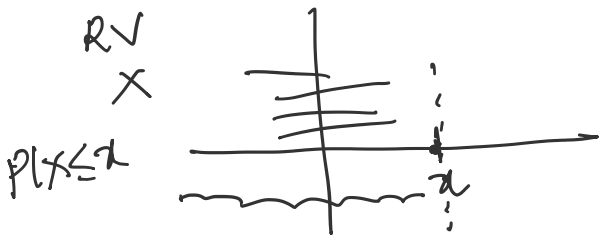


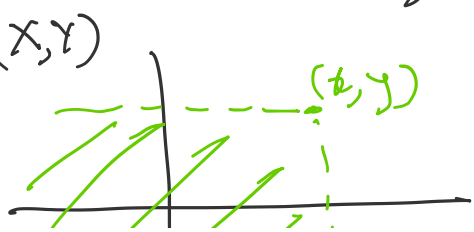


(X, Y) → Random vector

⇒ Joint CDF = $\mathbb{P}(X \le x, Y \le y)$

$\forall\ (x, y) \in \mathbb{R}^2$





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$(X_1, X_2 \dots X_n)$

$F_{X_1 X_2 X_3 \dots X_n}(x_1, x_2 \dots x_n) = \mathbb{P}(X_1 \le x_1, X_2 \le x_2 \dots, X_n \le x_n)$

Joint CDF

$x \le a, y \le y$

① $F_{XY}(x,y) = \mathbb{P}(X \le x, Y \le y)$

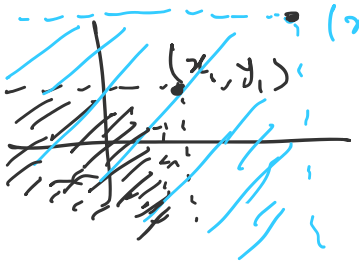
$\lim_{\substack{x \to \infty \\ y \to \infty}} F_{XY}(x,y) = 1$

$\lim_{\substack{x \to -\infty \\ y \to -\infty}} F_{XY}(x,y) = 0$

② (x_1, y_1) & (x_2, y_2) s.t $x_1 \le x_2$, $y_1 \le y_2$

$F_{XY}(x_1, y_1) \le F_{XY}(x_2, y_2)$

$\mathbb{P}(X \le x_1, Y \le y_1)$ $\quad$ $\mathbb{P}(X \le x_2, Y \le y_2)$



$A \subseteq B$

$P(A) \le P(B)$

③ $(X, Y) \longrightarrow F_{XY}$ } joint CDF

$\mathbb{P}(X \le x) \leftarrow F_X$ $\quad$ $F_Y \rightarrow \mathbb{P}(Y \le y)$

marginal CDFs

$F_{XY}(x,y) = \mathbb{P}(X \le x, Y \le y)$ }

$F_X(x) = \mathbb{P}(X \le x)$ } $= \lim_{y \to \infty} F_{XY}(x,y)$

$F_Y(y) = \mathbb{P}(Y \le y) = \lim_{x \to \infty} F_{XY}(x,y)$

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$\{X \le x \cap Y \le \infty\}$
$= \{X \le x\}$

Eg $\rightarrow$ (X, Y) $\quad X \rightarrow$ lifetime of CPU
$Y \rightarrow$ lifetime of battery

$\Rightarrow F_{XY}(x,y) = 1 - e^{-x} - e^{-y} + e^{-\left(x+y+\frac{xy}{2}\right)}$

$x, y \ge 0$

$\lim_{\substack{x\to\infty \\ y\to\infty}} F_{XY}(x,y) = 1$

$\lim_{x,y\to-\infty} F_{XY}(x,y) = \lim_{\substack{x\to 0 \\ y\to 0}} \left(1 - \underbrace{e^{-x}}_{\downarrow 1} - \underbrace{e^{-y}}_{\downarrow 1} + \underbrace{e^{-\left(x+y+\frac{xy}{2}\right)}}_{\downarrow 1}\right)$

$= 0$

P(Both comp. fail in 1st year)

$= P(X \le 1, Y \le 1) = F_{XY}(1,1)$

$= 1 - e^{-1} - e^{-1} + e^{-\left(1+1+\frac{1}{2}\right)}$

P(CPU fails in 1st year) $= P(X \le 1)$
$= F_X(1)$

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$$= \lim_{y\to\infty} F_{XY}(1,y) =$$

$$= \lim_{y\to\infty} 1 - e^{-1} - \underbrace{e^{-y}}_{\to 0} + \left(e^{-\left(1+y+\frac{y}{2}\right)}\right)$$

$$= 1 - e^{-1}$$

$F_{XYZ}(x,y,z)$. $F_X(x) = \lim_{\substack{y\to\infty \\ z\to\infty}} F_{XY}(x,y,z)$

Independence

Random vector $(X_1, X_2 \ldots X_n)$

Joint CDF $F_{X_1,X_2\ldots X_n}(x_1,x_2,\ldots x_n) \quad \forall (x_1,x_2\ldots x_n) \in \mathbb{R}^n$.

$X_1, X_2 \ldots X_n$ are independent if

$$F_{X_1,X_2\ldots X_n}(x_1,x_2\ldots x_n) = \prod_{i=1}^{n} F_{X_i}(x_i)$$

$\forall (x_1,x_2\ldots x_n) \in \mathbb{R}^n$

$\downarrow$

(X,Y) indep. $P(X\in S_1, Y\in S_2) = P(X\in S_1)\cdot P(Y\in S_2)$

$$F_{XY}(x,y) = 1 - e^{-x} - e^{-y} + e^{-(x+y)} \qquad x,y \geq 0$$

X, Y ind ? F_X , F_Y

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$$F_X(x) = \lim_{y\to\infty} F_{XY}(x,y) = 1 - e^{-x} \qquad x \geq 0$$

$$F_Y(y) = \lim_{x\to\infty} F_{XY}(x,y) = 1 - e^{-y} \qquad y \geq 0$$

$$F_{XY}(x,y) \overset{?}{=} F_X(x) \cdot F_Y(y)$$

$$(1-e^{-x})(1-e^{-y}) \qquad x, y \geq 0$$

$$= 1 - e^{-x} - e^{-y} + e^{-(x+y)}$$ ✓

X, Y ind.

$$P(X \leq 1, \ 1 \leq Y \leq 2) = P(X \leq 1) \cdot P(1 \leq Y \leq 2)$$

Discrete Random Vectors

(X, Y) Discrete RVs if supports of X & Y are countable.

Joint PMF $P_{XY}(x,y) \triangleq P(X = x, Y = y)$

$$\sum_{\substack{x \in S_X \\ y \in S_Y}} P_{XY}(x,y) = 1$$

Marginal PMF →

$$P_X(x) = P(X = x) \qquad x \in S_X$$

$$P_Y(y) = P(Y = y), \qquad y \in S_Y$$

$$P_X(x) = P(X = x) = \sum_{y \in S_Y} P(X = x, Y = y)$$

$P(X = x, Y = y) = P_{XY}(x,y)$

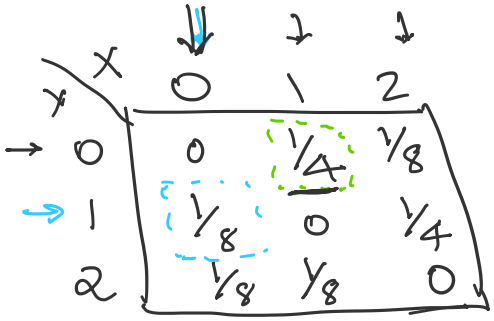
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$$\{X=x\} = \bigcup_{y \in S_Y} \{X=x \cap Y=y\} \rightarrow \text{disjoint union}$$

$$P(X=x) = \sum_{y \in S_Y} P(X=x, Y=y)$$

$$P_Y(y) = P(Y=y) = \sum_{x \in S_X} P_{XY}(x,y)$$

Eg (X,Y)



$P(X=1, Y=0) = 1/4$

$P(X=0, Y=1) = 1/8$

$P(X>Y) = 5/8$

$= P(X=1, Y=0) \rightarrow 1/4$
$+ P(X=2, Y=0) \rightarrow 1/8$
$+ P(X=2, Y=1) \rightarrow 1/4$

$P(X=0) = P(X=0, Y=0) + P(X=0, Y=1) + P(X=0, Y=2)$
$= 0 + 1/8 + 1/8 = 1/4$

$F_{XY}(1,1) = P(X \le 1, Y \le 1) = P(X=0, Y=0) + P(X=0, Y=1) + P(X=1, Y=0) + P(X=1, Y=1)$

Conditional PMF

$(X,Y) \rightarrow P_{XY}(x,y)$

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disc. RV

$$P_{X|Y}(x|y) \triangleq P(X=x \mid Y=y)$$

$$= \frac{P(X=x \cap Y=y)}{P(Y=y)}$$

$$= \frac{P_{XY}(x,y)}{P_Y(y)}$$

Fix y. Vary x

$$\sum_{x \in S_X} P_{X|Y}(x|y) = \sum_{x \in S_X} \frac{P_{XY}(x,y)}{P_Y(y)}$$

$$= \frac{1}{P_Y(y)} \cdot \sum_x P_{XY}(x,y) \to P_Y(y)$$

$$= 1.$$

Fix y. $\{P_{X|Y}(\cdot|y)\}_{x \in S_X}$ forms a PMF

$Y \backslash X$	0	1	2
0	0	1/4	1/8
1	1/8	0	1/4
2	1/8	1/8	0

$P(X=0 \cap Y=2)$

$$P(X=0 \mid Y=2) = \frac{P(X=0, Y=2)}{P(Y=2)} = \frac{P_{XY}(0,2)}{P_{XY}(0,2)+P_{XY}(1,2)+P_{XY}(2,2)}$$

$P(X=0) = 1/4$

$$P(X=1 \mid Y=2) = \frac{1/8}{1/4} = 1/2.$$

$$P(X=2 \mid Y=2) = 0$$

$$= \frac{1/8}{1/8+1/8+0} = \frac{1}{2}$$

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$$1 = P(X=0|Y=2) + P(X=1|Y=2) + P(X=2|Y=2)$$

Independence

(X,Y) ↘ $F_{XY}(x,y) = F_X(x) \cdot F_Y(y)$

↓

→ Discrete RV → $P_{XY}(x,y) = P_X(x) P_Y(y)$

$\forall\, x \in S_X$, $y \in S_Y$

(X,Y) Ind. ⇔ $P_{XY}(x,y) = P_X(x) P_Y(y)$

X, Y ind

$P(X=x|Y=y) = P_{X|Y}(x|y) = \dfrac{P_{XY}(x,y)}{P_Y(y)} = P_X(x) \rightarrow P(X=x)$

$P_Y(y) > 0$

$P(X \in B_1, Y \in B_2) = P(X \in B_1) \cdot P(Y \in B_2)$

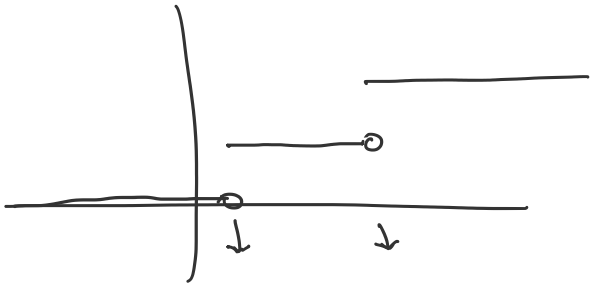
If (X,Y) $P_{XY}(x,y) = P_X(x) P_Y(y)$

$\forall\, x \in S_X$, $y \in S_Y$

For any $(z_1, z_2) \in \mathbb{R}^2$

$$F_{XY}(z_1, z_2) = F_X(z_1) \cdot F_Y(z_2)$$

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X = # visitors in a day

Y = # sales in a day

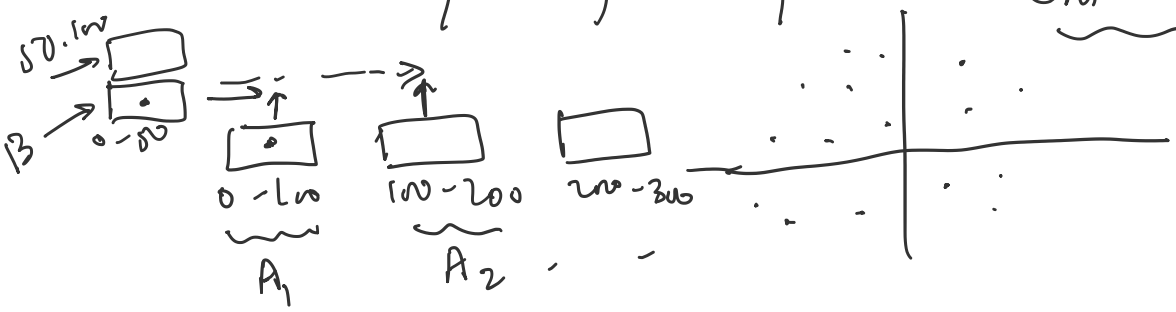
X	Y
1000	50
980	40
6000	75

Empirical dist.

Joint Dist. Model

Checking Model fit

Chi-squared test



X, Y ind.

$$P(X \in S_1, Y \in S_2) = P(X \in S_1) \cdot P(Y \in S_2)$$

$$P(0 < X < 100,\ 50 < Y < 60) = P(0 < X < 100) \cdot P(50 < Y < 60)$$

(B_1, B_2) X, Y ind.

$\Downarrow$

$\forall B_1, B_2$ $P(X \in B_1, Y \in B_2) = P(X \in B_1) \cdot P(Y \in B_2)$

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